FLUID DYNAMICS (AAE 2124)

MID-TERM EXAMINATION, SEPT. 2025

Exam Date: 18/09/2025

SOLUTION KEY

Q1 →

$$\tau = \mu \frac{du}{dy}$$

$$= 0.4 \times \frac{(0.7 - 0.3)}{0.2}$$

1M

$$\boxed{\tau = 0.8 \text{ Pa}}$$

Q2 →

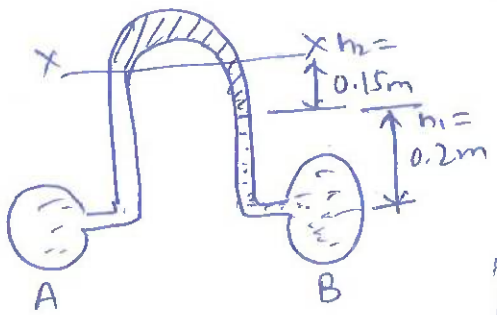
$$F_B = \rho_f g V$$

$$= 1000 \times 9.81 \times \frac{\pi}{4} \times 0.04^2 \times 0.1$$

1M

$$\boxed{F_B = 1.23 \text{ N}}$$

Q3 →



For right limb,

$$h_{Total} = h_B - h_1 s_1 - h_2 s_2$$

$$= 2.8 - 0.2 \times 1 - 0.15 \times 0.8$$

1M

$$\boxed{h_{Total} = 2.48 \text{m}}$$

1M

Q4 → below

Q5 → The body weight & buoyancy force create overturning moment.

1M

Q6 → Continuum Approach:

- continuous substance without any voids. 0.5M
- No discontinuity in property variation. 0.5M
- System size >> mean free path of molecules. 0.5M
- Example 0.5M

Q 7 →

Open System :

- Both mass & energy can cross the boundary — 0.5M
- Example — 0.5M

Closed System

- Mass can't cross boundary but energy can. — 0.5M
- Example — 0.5M

Isolated System

- Neither mass nor energy can cross the boundary. — 0.5M
- Example — 0.5M

Q 8 →

Stability of Immersed body :

Diagram — 1M

Stable - CG lies below B. - explanation — 1M

Neutrally stable - CG & B matches - explanation — 0.5M

Unstable - CG is above B - explanation — 0.5M

Q 9 →

Diagram — 1M

$$\frac{du}{dy} = \frac{v}{l} \quad \text{— 0.5M}$$

$$\tan d\beta = d\beta = \frac{da}{l} \quad \text{— 0.5M}$$

$$da = v \cdot dt$$

$$d\beta = \frac{v\,dt}{l}$$

$$\frac{d\beta}{dt} = \frac{v}{l} \quad \text{— 0.5M}$$

$$\therefore \boxed{\frac{d\beta}{dt} = \frac{du}{dy}} \quad \text{— 0.5M}$$

Q 10 →

Given:

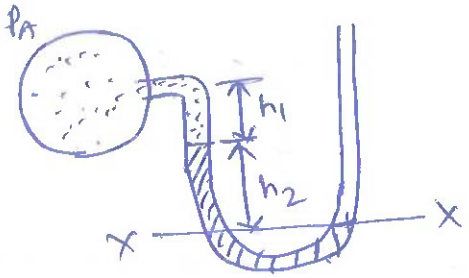
$s = 0.8$

$\rho_1 = 800\ kg/m^3$

$h_1 = 15cm = 0.15m$

$h_2 = 20cm = 0.2m$

$\rho_2 = 13,600\ kg/m^3$



1M

Above datum x-x, pressure ~~in the tank~~ is, balance eqn

1M

$$P_A + \rho_1 g h_1 + \rho_2 g h_2 = 0$$

$$P_A + 800 \times 9.81 \times 0.15 + 13,600 \times 9.81 \times 0.2 = 0$$

1M

$$\boxed{P_A = -27.86\ kPa}$$

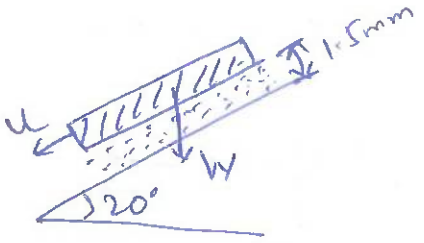
Q 11 →

Given: $dy = 1.5mm = 0.0015m$, $u = 0.2 m/s$,

$W = 392.4\ N$, $\rho = 700\ kg/m^3$

$A = 0.9 \times 0.9\ m^2$

$\nu = ?$



$$\tau = \frac{W \sin 20°}{A} = \frac{392.4 \times \sin 20°}{(0.9 \times 0.9)}$$

0.5M

0.5M

$$\tau = 165.69\ Pa$$

0.5M

$$\tau = \mu \frac{du}{dy}$$

$$165.69 = \mu \frac{0.2}{0.0015}$$

$$\mu = 1.24\ Pa \cdot s$$

0.5M

0.5M

$$\nu = \frac{\mu}{\rho} = \frac{1.24}{700}$$

$$\nu = 0.00178\ m^2/s$$

0.5M

Q 12 →

Continuity Equation

Diagram

1M

Net outflow in x-direction = outflow in x-dirn − inflow in x-dirn

$$= \left[eu + \frac{\partial (eu)}{\partial x} dx \right] dy\, dz - eu\, dy\, dz$$

$$= \frac{\partial}{\partial x} (eu)\, dx\, dy\, dz$$

Total net outflow for the element =

$$\left[\frac{\partial}{\partial x} (eu) + \frac{\partial}{\partial y} (ev) + \frac{\partial}{\partial z} (ew) \right] dx\, dy\, dz$$

1M

Time rate of change of mass inside the element,

$$= -\frac{\partial e}{\partial t} dx\, dy\, dz$$

1M

Total net outflow = Time rate of change of mass

$$\boxed{\frac{\partial e}{\partial t} + \frac{\partial (eu)}{\partial x} + \frac{\partial (ev)}{\partial y} + \frac{\partial (ew)}{\partial z} = 0}$$

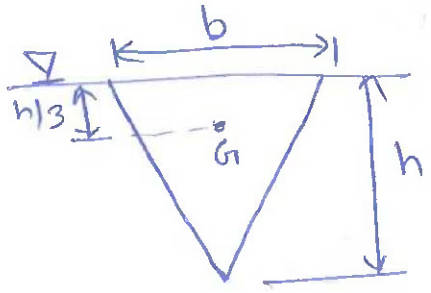
1M

Q 13 →

Given:

$b = 4m,\ h = 4m,\ s = 0.9$

$e = 900\ kg/m^3$



$$\bar{h} = \frac{h}{3} = \frac{4}{3} = 1.33\ m$$

0.5M

Total pressure, $F = egA\bar{h}$

$$= 900 \times 9.81 \times \frac{1}{2} \times 4 \times 4 \times 1.33$$

$F = 93.94\ kN$ — 1M (3)

$I_G = \frac{bh^3}{36}$ — 0.5M

$= \frac{4 \times (4)^3}{36}$

$I_G = 7.11\ m^4$ — 0.5M

$h^* = \frac{I_G}{A\bar{h}} + \bar{h}$ — 0.5M

$= \frac{7.11}{\frac{1}{2} \times 4 \times 4 \times 1.33} + 1.33$

$h^* = 2m$ — 1M